

Full-spectrum visible electro-optic modulator

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We report an on-chip high-speed visible-band electro-optic modulator that can operate over the full visible spectrum of 400–700 nm, with a record low $V_\pi \cdot L$ of 0.48, 0.25, and 0.17 V · cm at red, green, and blue wavelengths of 630, 520, and 450 nm, respectively, and an operation bandwidth of >20 GHz. © 2023 Optica Publishing Group under the terms of the Optica Open Access Publishing Agreement

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The visible spectral region underpins many important applications including sensing, optical clocks, 3D displays, and augmented/virtual reality (AR/VR). All of these applications rely crucially on precise control and efficient modulation of visible light. Recently, there has been significant interest in transferring these applications onto chip-scale platforms [1,2] that would offer great advantages in size, power, functionality, and design flexibility. However, development of chip-scale electro-optic modulators (EOMs) in the visible band remains fairly limited [3–5]. Here, we demonstrate an on-chip lithium niobate (LN) EOM that can operate over the full spectrum covering the entire visible band from 400 to 700 nm. Thin-film LN EOMs have attracted significant interest recently [6–13], the majority of which have been focused on the telecom band in response to the demand from data communication. Here, we show that the visible-band EOM exhibits record high modulation efficiency with $V_\pi \cdot L$ as low as 0.48, 0.25, and 0.17 V · cm at wavelengths of 630, 520, and 450 nm, respectively, which are the smallest ever reported for LN traveling-wave EOMs developed to date [3–13].

Figure 1(a) shows a fabricated EOM that consists of a pair of 3 dB multimode interference (MMI) couplers, an 8-mm-long phase modulation section operating in the push-pull fashion, and a spatial-mode filter (SMF) section placed in the front. The devices are made on a 300-nm-thick x-cut LN-on-insulator wafer, partially etched down by 180 nm. The modulator waveguide has a width of 1 μm and a waveguide-electrode gap (WEG) of 0.8 μm, with an electrode spacing of 2.6 μm, considerably smaller than other LN EOMs [3–13] in order to enhance modulation efficiency. The electrodes contact directly with the LN layer to improve the optical-microwave mode overlap. For the EOM to operate over the full visible spectrum, the MMI coupler is designed to be broadband with a transmission of 43.3%, 49.8%, and 42.8% at wavelengths of 400, 550, and 700 nm, respectively [Fig. 1(b)]. The SMF is a

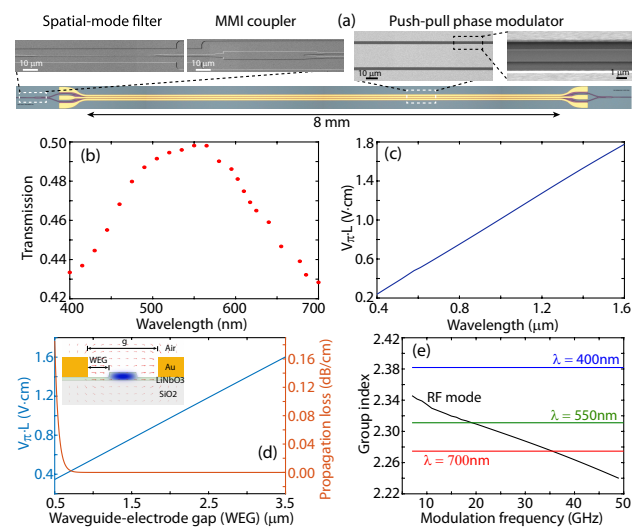


Fig. 1. (a) Image of a fabricated LN Mach-Zehnder EOM (MZM). (b) Numerically simulated transmission spectrum of the 3 dB MMI coupler. (c) Simulated wavelength dependence of $V_\pi \cdot L$ for a push-pull MZM with a WEG of 0.8 μm. (d) Simulated $V_\pi \cdot L$ (blue) and propagation loss (red) as a function of WEG at the wavelength of 630 nm. Inset shows the cross section of the modulator waveguide. (e) Simulated group index of the microwave (black) and the optical (color) modes. (c)–(e) are simulated with the finite-element method, and (b) is simulated with the finite-difference time-domain method.

tapered waveguide to cut off higher-order guided modes to ensure single-mode operation of the EOM.

For an LN phase modulator with a length of L , the induced phase shift at an optical wavelength of λ is given by $\delta\varphi \approx \frac{\pi}{\lambda} n_e^3 r_{33} \frac{VL}{g} \rho$, where n_e and r_{33} are the refractive index and electro-optic coefficient of LN, V is the voltage applied to the driving electrodes with a spacing of g in between, and ρ is the spatial overlap factor between the optical and microwave guided modes. Compared with the telecom band, operating in the visible band is beneficial in three ways. First, $\delta\varphi$ scales inversely with λ , resulting in a nearly linear wavelength dependence of $V_\pi \cdot L$ [Fig. 1(c)]. Second, $\delta\varphi$ scales inversely with g . The optical mode is more confined with decreased λ , leading to a smaller allowable WEG and waveguide width (and, thus, smaller g). As shown in Fig. 1(d), a waveguide width of 1 μm enables a WEG as small as <0.5 μm without introducing much extra loss, leading to $g < 2$ μm. Third,

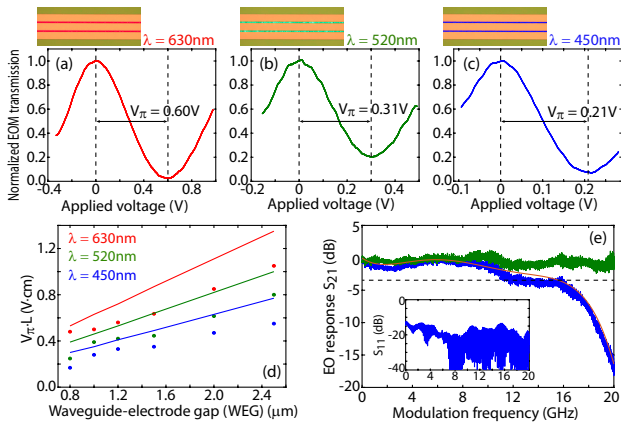


Fig. 2. (a)–(c) Measured EOM transmission as a function of applied voltage, at wavelengths of (a) 630 nm, (b) 520 nm, and (c) 450 nm, respectively. The insets show the device with laser input. (d) Recorded WEG dependence of $V_\pi \cdot L$, with experimental data shown as dots and theory shown as lines. (e) Recorded EO response S_{21} (blue) of the EOM. The red curve shows the response of the optical detector, and the green curve shows the S_{21} after factoring out the detector response. The inset shows the recorded S_{11} of the EOM, where the fine oscillation is due to reflection of RF connectors used in the testing setup.

r_{33} increases with decreased λ [14]. All of these factors contribute to enhancing the modulation efficiency. For high-speed operation, the EOM is designed to have a small group index mismatch between the microwave and optical modes over a frequency range $> 50\text{ GHz}$ [Fig. 1(e)].

Figures 2(a)–2(c) show the modulation performance of the EOM. The device exhibits a V_π of 0.60 V, 0.31 V, and 0.21 V, respectively, at the red, green, and blue wavelengths of 630, 520, and 450 nm, which corresponds to a $V_\pi \cdot L$ of 0.48, 0.25, and 0.17 V·cm. The extinction ratio (ER) is measured to be 16 dB at 630 nm [Fig. 2(a)]. The ER decreases to 7 dB and 12 dB at 520 and 450 nm [Figs. 2(b) and 2(c)], respectively, which is dominantly due to the green and blue FP lasers (Thorlabs, LP-520 and LP-450) with a poor polarization ER of $\sim 10\text{ dB}$ that interferes with the ER characterization. The recorded WEG dependence of $V_\pi \cdot L$ [Fig. 2(d)] shows a slightly better performance than the theoretical expectation, which is likely due to a smaller fabricated WEG than the designed WEG. Figure 2(e) shows that the EOM exhibits a 3 dB bandwidth of 16 GHz (blue curve). This value is simply limited by the frequency response of the optical detector (Newport, 1544-B) [Fig. 2(e), red curve]. By factoring out the detector response, the EOM itself exhibits a 3 dB bandwidth $> 20\text{ GHz}$ [Fig. 2(e), green curve]. The insertion loss of the EOM is measured to be 6.8 dB at 630 nm, which is primarily attributed to the fabrication imperfections of the SMF and the MMI couplers.

Figure 3 compares the EOM with state-of-the-art LN EOMs as well as high-speed monolithic Si and InP EOMs. These three types of EOMs are very promising for practical implementation compared with others. Clearly, this device exhibits the smallest $V_\pi \cdot L$ that has ever been reported for LN EOMs developed to date. It is even smaller than Si and InP EOMs, while the latter two can only work in the telecom band. The demonstrated full-spectrum EOM with record performance achieves a key step toward energy-efficient and high-speed visible photonics, opening up a great avenue toward chip-scale miniaturization and integration of versatile functionalities in sensing, atomic clocks, AR/VR, etc., on the promising thin-film LN platform.

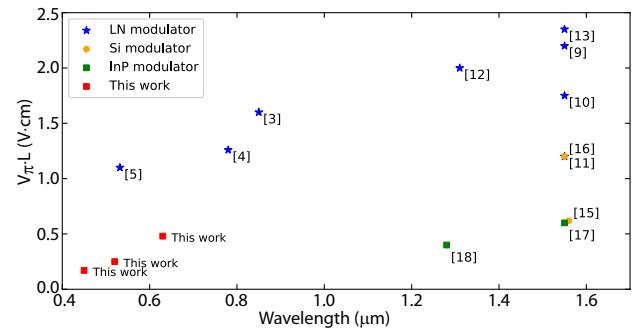


Fig. 3. Comparison of the EOM with state-of-the-art LN [3–5,9–13], silicon [15,16], and InP [17,18] Mach-Zehnder modulators.

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Data availability. Data are available upon reasonable request.

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